FOUNDATIONS OF GENERATIVE AI (ICT4403)

Assignment 2: Autoregressive Text Generation System — Comparative Study

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Assignment Title	Assignment 2 - Autoregressive Text Generation System
System Title	Comparative Word-Level Autoregressive Text Generation on Lewis Carroll's Alice in Wonderland using Simple RNN, BiRNN, LSTM, GRU
Dataset & Source	Alice's Adventures in Wonderland (Lewis Carroll) | https://www.gutenberg.org/files/11/11-0.txt
GitHub Repository	https://github.com/Saksham-Gupta-GH/fgai-autoregressive-text-generation-alice

1. System Overview & Problem Statement

Four recurrent neural language models (Simple RNN, BiRNN with forward-backward final state concatenation $[h_n[-2], h_n[-1]]$, LSTM, and GRU) were trained under strictly controlled conditions on Alice's Adventures in Wonderland. To eliminate data leakage, the corpus is partitioned chronologically (80% Train, 10% Val, 10% Test) and the vocabulary (1,300 tokens, `min_freq=2`) is fitted exclusively on training tokens. Context sequence length is 6 tokens. Held-out test set has an 11.17% `<unk>` token rate.

2. Quantitative Evaluation Summary (Best Epoch & Test Set)

Architecture	Parameters	Best Ep	Val Loss	Val PPL	Val Top-1	Test Loss	Test Top-1	Test Top-5
Simple RNN	798,484	4	4.0741	58.80	19.73%	4.1584	20.80%	45.28%
BiRNN	1,262,868	2	4.2804	72.27	18.66%	4.3192	18.47%	42.26%
LSTM	1,193,236	5	3.9903	54.07	21.87%	4.0576	23.29%	47.23%
GRU	1,061,652	3	4.0353	56.56	20.55%	4.0900	22.53%	46.51%

3. Cross-Model Temperature Analysis (Averaged Over Models and Seed Prompts, $n = 3$ per cell)

T	Distinct-1	Distinct-2	Trigram Repetition	Longest Loop	Empirical Quality Observation
0.2	0.517	0.674	23.7%	0–6 words	Strict syntax; repetitive looping traps across models (GRU 52.5%)
0.5	0.678	0.900	3.0%	0 words	Optimal empirical trade-off: high diversity with fluent story dialogue; LSTM and GRU comparable
1.0	0.855	0.992	0.0%	0 words	High lexical richness; slight syntactic drift across longer clauses
1.5	0.916	1.000	0.0%	0 words	High entropy; syntactic collapse into ungrammatical token sequences

4. Conclusion & Model Comparison

- LSTM vs. GRU & UNK Nuance: LSTM leads across next-word prediction metrics (Val Loss 3.9903, PPL 54.07, Test Top-1 23.29%), with GRU performing closely behind (Val Loss 4.0353, Test Top-1 22.53%) with 11% fewer parameters. Test targets include 11.17% `<unk>` tokens due to train-only vocabulary fitting.
- Qualitative Consistency & Seed Origin: Seed 1 comes from the training text, Seed 2 from held-out test text, and Seed 3 from validation text. Consistency was assessed qualitatively: at $T = 0.5$, LSTM and GRU more frequently generate Carroll-style dialogue tags ('the march hare said'), though all models drift across multi-clause sequences without sustaining a persistent scene.
- Single-Seed Limitation & BiRNN: `SEED=42` ensures strict comparability; the 0.045 loss gap between LSTM and GRU falls within single-run variance margins. BiRNN's forward-backward state concatenation feeds complete context, but its 1,262,868 parameters overfit by epoch 2 (val loss 4.2804).
- Temperature Sampling: $T = 0.5$ provides the optimal trade-off between lexical diversity (Distinct-1: 0.60–0.73) and phrase stability, achieving minimal trigram repetition (LSTM 5.0%, GRU 0.0%, BiRNN 1.0%, Simple RNN 6.1%).